



USC

PHYS 513 Applications of quantum computing

Units: 4

Term—Day—Time: Two 100 minutes classes per week

Location: DMC 207

Instructor: Prof. Rosa Di Felice and Prof. Aiichiro Nakano

Office: SSC 215D (RDF), VHE 610 (AN)

Office Hours: upon appointment

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Course Description

This class will give a one-semester graduate-level training on available quantum computing hardware and the respective functional principles, available software and **selected** algorithms. We will focus on gate-model IBM and adiabatic D-Wave devices and their **applications mostly in chemistry and materials sciences**. Students are welcome to deepen applications in other fields for the final project.

The students will have the opportunity to program quantum computers and run sample problems. Some classes will include hands-on tutorials, using own laptops. Information for obtaining credentials on quantum computers will be given in class.

Learning Objectives

Quantum hardware that is now available to users on the cloud include adiabatic quantum optimizers and gate-logic devices. The different principles of operation will be introduced at a practical level, for the purpose of computer programming rather than theoretical foundations/development.

After the first introductory week, the course will split in two parts, separately on gate-logic quantum computers (about 10 out of 14 weeks) and adiabatic quantum computers (about 4 out of 14 weeks).

Gate-logic quantum computing.

- The existing IBM devices will be characterized, in terms of qubit types and qubit connectivity.
- Existing software will be illustrated, in particular Qiskit.
- We will consider classes of problems that may be solved on and take advantage from these quantum devices
- We will teach the basics on quantum chemistry and quantum dynamics, at the level to prepare the students for programming these classes of problems on the existing hardware exploiting the available software, as well as future-generation hardware and software.
- We will explain and use few quantum-classical hybrid algorithms for molecular energy estimation, including methods for the reduction of qubit counts and circuit depths.
- A section of the course will be devoted to quantum dynamics on quantum computers, taught by Prof. Nakano in weeks 13-14.

Programming in Qiskit will not be taught systematically, but students will have to self-learn for homework, midterm and final project. Some examples of available tutorials will be presented in class. Qiskit is a python-based software, python coding is an essential co-requisite for this course.

Adiabatic quantum computing.

- After reviewing the quantum adiabatic theorem, the D-Wave available devices will be outlined.
- We will give an overview of the Ocean software.
- We will consider classes of problems that may be solved on and take advantage of adiabatic quantum optimizers.
- We will teach the basics on optimization problems focusing on multiple linear regression, implementation of quantum chemistry on D-Wave and mapping of graphs and networks onto quadratic unconstrained binary optimization (QUBO) forms.
- We will explain minor embedding to bypass the limited qubit connectivity.

Other. Introduction to different hardware and software packages: In-Quanto by Quantinuum, PennyLane by Xanadu, CUDA-Q by NVIDIA.

Eventually, the students will be able to program and run at least a couple of the above problem types on the available quantum hardware.

Recommended preparation: introductory quantum mechanics and quantum computing; python coding.

Co-requirement: EE 520 (or PHYS 558A).

Course Notes

This course will comprise lectures, homework, a midterm exam and a student research project. The course will ordinarily be taken for a letter grade. Documents, including lecture notes, homework assignments, and additional readings, will be distributed online via the course Brightspace site.

Technological Proficiency and Hardware/Software Required

Prior knowledge of python, Jupiter and notebooks is an advantage, or proficiency with these tools will have to be acquired during the course.

Student projects and tutorials will involve programming.

Familiarity with a variety of operating systems, including Linux, will greatly help.

Required Readings and Supplementary Materials

Lecture notes and research papers or other handouts, available via Brightspace.

For quantum chemistry: "Introduction to computational chemistry", by Frank Jensen (Third edition);

"Molecular Electronic-Structure Theory", by Trygve Helgaker, Poul Jørgensen, Jeppe Olsen (Wiley).

Description and Assessment of Assignments

The coursework will include 6 homework assignments, a midterm assignment and a student final project. The assignments will draw on the material learned in class. Topics are specified in the weekly breakdown.

Description of Student Project

The project will be based on programming and running on a noisy intermediate-stage quantum (NISQ) computing machine. Codes available on GitHub, QISKIT or other similar platforms can be used as starting points.

It can be one of: (1) reproducing a published research paper; (2) a piece of original research; (3) a review-type project (no coding/running). The latter will only reach a maximum of 75% of the available final grade.

The subject of the project will be agreed on between the student and the course instructor(s); the instructor(s) will suggest suitable sources for the selected topic, and will meet with each student in the second half of the semester to give guidance on the project. The students will have access over the cloud to NISQ machines from IBM and D-Wave, and potentially other devices.

Project deliverables will include a written report and an oral presentation in class. Of the 50% it contributes to the final grade (37.5% for option 3), the report and oral presentation are worth half and half, respectively. The report is expected to be about 4-6 pages (preprint style), including figures and references (at least 1 figure, at least 5 references). The presentation is expected to be 20-30 minutes.

Typical project topics: ground-state energy estimation of small molecules on quantum devices and classical simulators with the variational quantum eigensolver and other quantum or hybrid algorithms; programming simple circuits; analyzing qubit coherence; optimization problems on materials or biological data, error correction strategies; natural language processing; quantum machine learning and neural networks.

Grading Breakdown

Course Element	% of Grade (final options 1 and 2)	% of Grade (final option 3)
Problem Sets (6)	18%	18%
Midterm Exam	32%	32%
Student Project (paper and presentation)	50%	37.5%
TOTAL	100% -- max A	87.5% -- max B+

Assignment Submission Policy

Problem sets will be submitted through Brightspace on the assigned date. Late homework will not be accepted. After the midterm exam, students will choose their project topics; the presentations will be scheduled for the last week of class, and the project reports will be due on the date of the scheduled final exam (Tuesday, May 12, 2025, 1pm).

Grading Timeline

Strong effort will be made to grade and return homework a week after it is received.

Participation

Active participation is encouraged. Examples include, but are not limited to: asking questions, making comments, proposing alternative explanations to topics presented in class, suggesting additional topics for the table of contents, promoting small-group study. While no specific grade is assigned to Participation, it will be considered by the instructors to assign the final grade, in particular in borderline cases (e.g., at the threshold between two letters).

Attendance

Total number of lectures in the course: 27. This is the result of 2 lectures per week for 15 weeks, minus two lectures for students' presentations in the last week, minus 1 lecture for the midterm exam. The midterm exam is take home, but one class will be skipped to allow students to work on it full-time during the class time, before the submission deadline.

No portion of the grade will be awarded for class attendance. Non-attendance can be the basis for lowering the grade: specifically, 60% of in-person attendance (15 lectures) is required to get the full credit accumulated with homework, midterm and final exam. Each missed class bringing the total number of attended classes below this threshold will result in lowering the final grade by 2%. Make up opportunities are not offered.

Exceptions. Student athletes with approved Travel Request Letters are an exception. Students who give advance notice of religious observation are an exception. These students will be offered the opportunity to make up for their absences: (1) by attending on zoom; (2) by submitting a short video presentations of the topics presented in class.

Statement for observance of religious holidays

USC's policy grants students excused absences from class to observe religious holidays: <http://orl.usc.edu/life/calendar/absences/>. In this case, please contact your instructor in advance to agree on alternative course requirements.

Course Schedule: A Weekly Breakdown (approximate)

		Topics	Readings	Deliverables
Week 1 Di Felice	1/12-1/16, 2026	Fundamentals of gate-logic quantum computation and adiabatic quantum computation; differences between the two paradigms, and their equivalence.	TBA	
Week 2 Di Felice	1/19-1/23, 2026	- Hardware in IBM. Principles of superconducting qubits (transmons). - Topology and connectivity of IBM cloud devices. Quantum logic gates on single and multiple qubits. Circuit depth.	TBA	HW1 due (basics)
Week 3 Di Felice	1/26-1/30, 2026	- Software on the IBM quantum platform: QISKIT, its modules and plugins. Tutorials and demonstrations of QISKIT. - Access to IBM quantum platform through USC-QIC. - Hands-on demos.	IBM software manuals, additional lecture notes and research/review papers	
Week 4 Di Felice	2/2-2/5, 2026	Characterization of classes of problems that may be solved advantageously using quantum logic circuits: quantum chemistry, quantum dynamics, optimization.	Extracts from books will be distributed via Brightspace	HW2 due (use of QISKIT to program simple circuits)
Week 5 Di Felice	2/9-2/13, 2026	Quantum chemistry. The many-body problem for the electronic structure. Methods to solve the electronic structure of solids and molecules: HF, DFT, configuration interaction, active space, coupled cluster, DMRG, benchmarks. The molecular Hamiltonian.	Textbooks, research/review papers TBA	
Week 6 Di Felice	2/16-2/20, 2026	- The quantum phase estimation (QPE) algorithm for ground-state energy estimation. What has been done with QPE for quantum chemistry and materials science. - Molecular ground-state energy estimation in second quantization: the hybrid classical-quantum variational quantum eigensolver (VQE). - Going beyond VQE for excited state properties of materials (electron energy spectrum, excitons, photoluminescence, etc.): active space downsizing, environment, circuit optimization, other.	Research/review papers, IBM manuals	HW3 due (computational electronic structure problems with available classical software packages)
Week 7 Di Felice	2/23-2/27, 2026	- Preparation of the initial state for VQE: new algorithmic approaches. - Molecular ground-state energy estimation in second quantization: the sample-based quantum diagonalization (SQD). Presentation of published articles on SQD application: reaching the limit of 100+ qubits.	Research/review papers, lecture notes	
Week 8 Di Felice	3/2-3/6, 2025	- Molecular ground-state energy estimation in first quantization. - Beyond IBM gate-model devices and software packages: Quantinuum, Xanadu, Google, NVIDIA CUDA-Q. - Resource estimation: pyLIQTR.	Lecture notes, research/review papers	

Week 9 Di Felice	3/9-3/13, 2025	Materials science. - Spin chains (to be developed in weeks 13-14) - Multi-step computational protocol for industrial problems – portable to drug discovery.	Lecture notes and research/review papers	
Spring break 3/15-3/22, 2026				
Week 10 Di Felice	3/23-3/27, 2025	Midterm exam 3/24/2026: quantum chemistry, VQE, SQD. The adiabatic theorem of quantum mechanics and its implementation on D-Wave devices. The D-Wave hardware: superconducting qubits, magnetic field coupling, Chimera and Pegasus graphs. Minor embedding.	D-Wave manuals, research/review papers	HW5 due (gate-model quantum software)
Week 11 Di Felice	3/30-4/3, 2025	Programming D-Wave: the Ising Hamiltonian. Limitations of connectivity and “gauge”. Different hardware generations.	Lecture notes, book chapters TBA	Choice of student project
Week 12 Di Felice	4/6-4/10, 2025	Multiple linear regression on D-Wave. Quantum chemistry on D-Wave. Graphs and networks on D-Wave. Connection to materials science.	Lecture notes and research/review papers	HW6 due (D-Wave)
Week 13 Nakano	4/13-4/17, 2026	Quantum dynamics: quantum many-body dynamics on NISQ computers, quantum compiler optimization.	Lecture notes, research/review papers	
Week 14 Nakano	4/20-4/24, 2026	Quantum dynamics: quantum-circuit simulators: Schroedinger vs. Feynman solvers.	Lecture notes, research papers	HW4 (quantum dynamics)
Week 15	4/27-5/1, 2025	Presentation of student projects.		Final projects: oral presentations in class. Slides submitted through Brightspace.
FINAL	5/12/2025	Tuesday, May 12, 2025, 11am-1pm. Written report on final project. The written report should be in the format of a short manuscript, with Introduction, Methods, Results and Analysis, Summary and Conclusions. A latex template will be provided.		Project reports due by 1pm on 5/12/2026

IMPORTANT DATES

- Beginning of classes – Monday, January 12, 2026
- Martin Luther King Day (University holiday) – Monday, January 19, 2026

- Last day to add classes – Friday, January 30, 2026
- Last day to drop without a “W” and receive a refund – Friday, January 30, 2025
- Last day to change to pass/no pass – Friday, January 30, 2026
- President’s Day (University holiday) – Monday, February 16, 2026
- Last day to withdraw without a mark of “W” on the transcript or change pass/no pass to letter grade – Friday, February 27, 2026
- Spring recess (University holiday) – March 15-22, 2026
- Midterm exam – Tuesday March 24, 2026
- Last day to drop with a mark of "W" – Friday, April 10, 2026
- Last day of class – Friday, May 1, 2026
- Study days – May 2-6, 2026
- Final Exam – Tuesday, May 12, 2026, 11am-1pm

Statement on Academic Conduct and Support Systems

Academic Conduct:

Plagiarism – presenting someone else’s ideas as your own, either verbatim or recast in your own words – is a serious academic offense with serious consequences. Please familiarize yourself with the discussion of plagiarism in SCampus in Part B, Section 11, “Behavior Violating University Standards”, <https://policy.usc.edu>. Other forms of academic dishonesty are equally unacceptable. See additional information in SCampus and university policies on scientific misconduct, <https://policy.usc.edu/research-and-scholarship-misconduct/>.

Use of AI tools:

The use artificial intelligence (AI)-powered programs is allowed to help students with assignments. You should be aware that AI tools may present incorrect information, biased responses, and incomplete analyses; thus they are not prepared to produce codes and text that meet the standards of this course, without further student control. You must cite any AI-generated material (e.g., codes, text, images, etc.) included or referenced in your work. Using an AI tool to generate content without proper attribution will be treated as plagiarism and reported to the Office of Academic Integrity.

Support Systems:

Counseling and Mental Health - (213) 740-9355 – 24/7 on call
studenthealth.usc.edu/counseling

Free and confidential mental health treatment for students, including short-term psychotherapy, group counseling, stress fitness workshops, and crisis intervention.

National Suicide Prevention Lifeline – 988 Lifeline – 24/7 on call
<https://988lifeline.org>

Relationship and Sexual Violence Prevention and Services (RSVP) - (213) 740-9355(WELL), press "0" after hours – 24/7 on call

<https://sites.usc.edu/clientservices/>

Free and confidential therapy services, workshops, and training for situations related to gender-based harm.

Office of Equity and Diversity (OED)- (213) 740-5086 | Title IX – (213) 821-8298

equity.usc.edu

Information about how to get help or help someone affected by harassment or discrimination, rights of protected classes, reporting options, and additional resources for students, faculty, staff, visitors, and applicants. The university prohibits discrimination or harassment based on the following *protected characteristics*: race, color, national origin, ancestry, religion, sex, gender, gender identity, gender expression, sexual orientation, age, physical disability, medical condition, mental disability, marital status, pregnancy, veteran status, genetic information, and any other characteristic which may be specified in applicable laws and governmental regulations. The university also prohibits sexual assault, non-consensual sexual contact, sexual misconduct, intimate partner violence, stalking, malicious dissuasion, retaliation, and violation of interim measures.

Reporting Incidents of Bias or Harassment - (213) 740-5086 or (213) 821-8298

usc-advocate.symplicity.com/care_report

Avenue to report incidents of bias, hate crimes, and microaggressions to the Office of Equity and Diversity | Title IX for appropriate investigation, supportive measures, and response.

The Office of Disability Services and Programs - (213) 740-0776

dsp.usc.edu

Support and accommodations for students with disabilities. Services include assistance in providing readers/notetakers/interpreters, special accommodations for test taking needs, assistance with architectural barriers, assistive technology, and support for individual needs.

USC Support and Advocacy - (213) 821-4710

uscsa.usc.edu

Assists students and families in resolving complex personal, financial, and academic issues adversely affecting their success as a student.

USC Emergency - UPC: (213) 740-4321, HSC: (323) 442-1000 – 24/7 on call

dps.usc.edu

Emergency assistance and avenue to report a crime. Latest updates regarding safety, including ways in which instruction will be continued if an officially declared emergency makes travel to campus infeasible.

USC Department of Public Safety - UPC: (213) 740-6000, HSC: (323) 442-120 – 24/7 on call

dps.usc.edu

Non-emergency assistance or information.